Computational Economics

Lecture: Comparing VFI, PFI, GPI

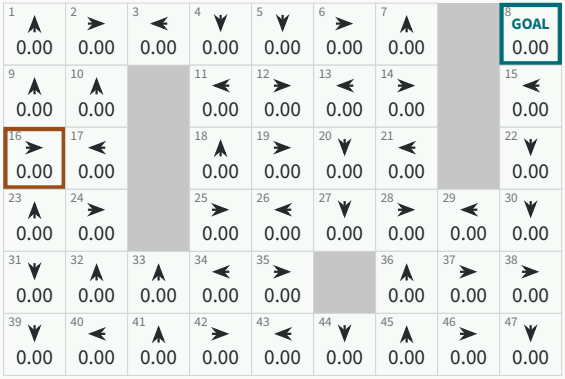
Yaolang Zhong yaolang.zhong@e.u-tokyo.ac.jp
Graduate School of Economics, The University of Tokyo

Lecture Overview

- **Value function iteration (VFI)**
  - Greedy policy selection and one evaluation update
- **Policy function iteration (PFI)**
  - Full policy evaluation and policy improvement
- **Generalized policy iteration (GPI)**
  - Choosing the number of evaluation updates
  - Modified policy iteration with $k = 3$

Demonstration pages contain embedded animations. Open in desktop Acrobat Reader and use the controls beneath each figure.

Initial Policy and Value Estimates



Fixed color scale for the current value estimate

Same initialization

$$V_0(s) = 0 \quad \text{for every cell.}$$

A fixed random deterministic policy assigns one action to each nonterminal cell.

- **Black arrow**: current action.
- **Red arrow**: action changed in this improvement step.
- **Cell fill and number**: current value; darker blue means more negative.
- **Orange outline**: start.
- **Teal outline**: goal.

Policy Improvement and Policy Evaluation

Use the same transition probabilities $P(s' \mid s, a)$ and discount $\beta = 0.9$ as before.

Policy improvement: compare all four actions at every cell using the same value array.

$$Q_V(s,a) = r(s,a) + \beta \mathbb{E}[V(s') \mid s, a], \qquad \sigma_{\text{new}}(s) \in \arg\max_{a \in \mathcal{A}(s)} Q_V(s,a).$$

One policy evaluation sweep: hold the policy fixed and update every cell once.

$$V_{\text{new}}(s) = r(s, \sigma(s)) + \beta \mathbb{E}[V_{\text{old}}(s') \mid s, a = \sigma(s)].$$

Hold the policy fixed and keep sweeping until convergence:

$$V_{j+1} = T_\sigma V_j, \qquad V_j = T_\sigma^j V_0 \longrightarrow V^\sigma \quad \text{as } j \to \infty.$$

Full policy evaluation by linear algebra: solve the fixed-policy Bellman equations.

$$V^\sigma = r^\sigma + \beta P^\sigma V^\sigma, \qquad (I - \beta P^\sigma) V^\sigma = r^\sigma.$$

VFI: Algorithm

Step 0. Initialize. Same random policy and $V_0 = 0$. Let $i = 0$.

Step 1. Improve at every state. Hold V_i fixed:

$$\sigma_{i+1}(s) \in \arg\max_{a \in \mathcal{A}(s)} \{r(s,a) + \beta \mathbb{E}[V_i(s') \mid s, a]\}.$$

Step 2. Evaluate once. Hold σ_{i+1} fixed:

$$V_{i+1}(s) = r(s, \sigma_{i+1}(s)) + \beta \mathbb{E}[V_i(s') \mid s, a = \sigma_{i+1}(s)].$$

Equivalently, $V_{i+1} = TV_i$.

Step 3. Stop or repeat. Stop when $\|TV_{i+1} - V_{i+1}\|_\infty < 10^{-10}$; otherwise increment i and repeat Steps 1–2.

Return the final value and a greedy policy. This run takes **15 rounds: 15 improvements and 15 evaluation sweeps**.

VFI: Demonstration

Initialization

Update 0 / 31



Fixed color scale for the current value estimate

Play forward / pause; step backward / forward; first / last; adjust speed.

PFI: Algorithm

Step 0. Initialize. Same random policy and displayed $V = 0$. Let $i = 0$.

Step 1. Evaluate fully. Hold σ_i fixed and solve for all states together:

$$V^{\sigma_i}(s) = r(s, \sigma_i(s)) + \beta \mathbb{E}[V^{\sigma_i}(s') \mid s, a = \sigma_i(s)].$$

$$(I - \beta P^{\sigma_i}) V^{\sigma_i} = r^{\sigma_i}.$$

Step 2. Improve at every state. Hold the evaluated values fixed:

$$\sigma_{i+1}(s) \in \arg\max_{a \in \mathcal{A}(s)} \{r(s, a) + \beta \mathbb{E}[V^{\sigma_i}(s') \mid s, a]\}.$$

Step 3. Stop or repeat. If no action changes, return (V^{σ_i}, σ_i). Otherwise increment i and repeat Steps 1–2.

This run takes **12 rounds: 12 full linear solves and 12 improvements**. Each solve replaces evaluation to convergence; it is not one sweep.

PFI: Demonstration

Initialization

Update 0 / 25



Fixed color scale for the current value estimate

Play forward / pause; step backward / forward; first / last; adjust speed.

GPI: Modified Policy Iteration: Algorithm

Step 0. Initialize. Same random policy, $V_0 = 0$, and a chosen $k \geq 1$.

Step 1. Improve at every state. Use the current value array:

$$\sigma_{i+1}(s) \in \arg\max_{a \in \mathcal{A}(s)} \{r(s,a) + \beta \mathbb{E}[V_i(s') \mid s, a]\}.$$

Step 2. Evaluate exactly k times. Fix this policy and set $V_{i,0} = V_i$:

$$V_{i,j+1}(s) = r(s, \sigma_{i+1}(s)) + \beta \mathbb{E}[V_{i,j}(s') \mid s, a = \sigma_{i+1}(s)], \quad j = 0, \ldots, k-1.$$

Set $V_{i+1} = V_{i,k}$. Each inner update is a full simultaneous sweep.

Step 3. Stop or repeat. Use the same Bellman-residual criterion as VFI. Return the final value and a greedy policy.

With $k = 3$: **12 rounds, 12 improvements, 36 sweeps**. Finite-k MPI is an instance of GPI; $k = 1$ reproduces VFI.

GPI: Modified Policy Iteration: Demonstration

Initialization

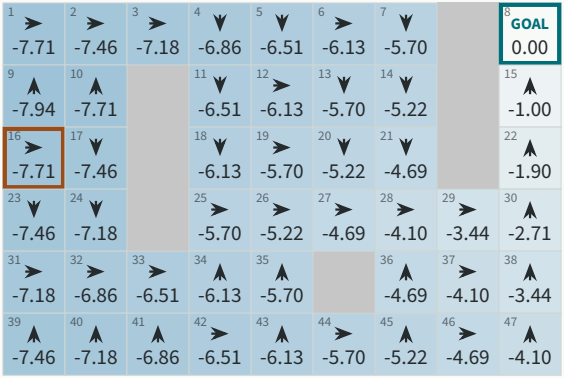
Update 0 / 49



Fixed color scale for the current value estimate

Play forward / pause; step backward / forward; first / last; adjust speed.

The Common Optimal Value



Fixed color scale for the current value estimate

Choosing the Amount of Evaluation

Method	Greedy selections	Evaluation sweeps	Linear solves
VFI	15	15	0
MPI, $k = 3$	12	36	0
PFI	12	0	12

Counts describe this maze and this initialization. A linear solve and a sweep have different costs.

- $k = 1$: VFI. More frequent greedy policy selection.
- Finite $k > 1$: MPI. More evaluation with each policy held fixed.
- Full evaluation between improvements: PFI.

All three reach the same optimal values. Ties may produce different optimal policies.